

THERMOGUARD™ TMS-450

Three-Phase Industrial Motor Thermal Protection & Monitoring System

Installation, Operation, Maintenance and Troubleshooting Manual

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NOTICE — This is fictional technical documentation created for FixRAG retrieval testing. It is not a real manufacturer's manual and is not a certified safety procedure.

Document Control & Table of Contents

Revision	Date	Description
1.0	Jan 2026	Initial fictional release
2.0	Feb 2026	Expanded diagnostics and maintenance
2.1	Mar 2026	Added testing scenarios and controlled appendix discrepancy

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Manual use

Read the safety section before hazardous work. Section references are included so technicians and retrieval systems can connect symptoms, procedures, specifications and warnings.

1. Product Overview

The fictional **ThermoGuard TMS-450** is a three-phase industrial motor thermal protection and monitoring controller for three-phase induction motors. It monitors motor temperature through TS1 and current through CT1, generates alarm indications, and operates a trip relay when configured protection criteria persist.

Primary functions

- Continuous temperature monitoring through TS1.
- Current monitoring through CT1.
- Configurable temperature alarm and trip thresholds.
- Configurable overload/current threshold and delays.
- Manual or supervised reset.
- POWER, STATUS, ALARM, FAULT and SENSOR indicators.
- Alarm and trip relay outputs.
- J1 configuration and J2 communication interfaces.

Environmental requirements

Parameter	Requirement
Operating ambient	–20 °C to +55 °C
Storage	–30 °C to +70 °C
Humidity	5–95% non-condensing
Location	Indoor industrial control enclosure
Ventilation	Keep controller ventilation path clear

Operating concept

In normal monitoring the controller samples TS1 and CT1 and compares them with configured limits. A persistent threshold condition produces an alarm; a qualifying trip condition operates K2 and records a protection fault.

2. Safety Information

DANGER — Three-phase power can cause fatal injury. Isolate energy, apply lockout/tagout and verify absence of hazardous voltage before opening or rewiring the enclosure.

WARNING — Unexpected motor startup can occur when control power is restored. Prevent automatic restart during troubleshooting.

WARNING — Motor surfaces may remain hot after a thermal trip. Allow safe cooling before physical inspection.

CAUTION — Do not use reset as a substitute for finding the underlying fault.

Required practices

- Use qualified personnel and site-approved PPE.
- Treat stored electrical energy as hazardous until discharged and verified safe.
- Keep rotating machinery guarded.
- Do not defeat protective interlocks or short sensor inputs.
- Record measured values and configuration before replacing components.
- Energized testing is permitted only when specifically required by the applicable procedure and performed by appropriately qualified personnel.

Safe diagnostic sequence

- 1 Stop the motor when practical.
- 2 Isolate energy and apply lockout/tagout.
- 3 Verify absence of hazardous voltage.
- 4 Inspect and diagnose without defeating safeguards.
- 5 Record findings.
- 6 Restore covers/guards before authorized re-energization.

3. System Components

Label	Component	Function
TB1	Power terminal block	Controller supply input
TB2	Sensor terminal block	TS1 connection
TB3	Relay terminal block	K1 alarm and K2 trip contacts
TB4	CT terminal block	CT1 measurement
J1	Configuration interface	Local service/configuration
J2	Communication port	Fictional RS-485 service interface
LED1	POWER	Supply indication
LED2	STATUS	Normal monitoring indication
LED3	ALARM	Alarm indication
LED4	FAULT	Protection trip indication
LED5	SENSOR	Sensor diagnostic indication
K1	Alarm relay	Alarm output
K2	Trip relay	Protective trip output
TS1	Temperature sensor	Motor temperature input
CT1	Current transformer	Motor-current input

Component notes

The labels above are fictional identifiers for this manual. Maintain ventilation clearance and protect TS1/CT1 wiring from mechanical damage and unsuitable environmental conditions.

4. Technical Specifications

Specification	Primary value
Supply voltage	100–240 VAC, 50/60 Hz
Maximum monitored current	80 A RMS
Temperature sensor range	–20 °C to +150 °C
Default alarm temperature	95 °C
Default trip temperature	110 °C
Operating temperature	–20 °C to +55 °C
Storage temperature	–30 °C to +70 °C
Response time	≤250 ms measurement update
K1 relay rating	250 VAC, 3 A resistive maximum
K2 relay rating	250 VAC, 3 A resistive maximum
Digital input	24 VDC nominal
Communication	J2 fictional RS-485 service protocol
Enclosure	IP42
Weight	0.62 kg
Dimensions	145 × 95 × 62 mm

Protection parameters

Parameter	Range	Default
Temperature alarm	50–130 °C	95 °C
Temperature trip	60–140 °C	110 °C
Overload threshold	50–120% of configured current	105%
Alarm delay	0–60 s	5 s
Trip delay	1–120 s	10 s
Reset mode	Manual / supervised	Manual
CT ratio	1:1 to 100:5 equivalent	20:1

NOTICE — The Appendix on page 26 intentionally lists a conflicting maximum monitored current of 75 A RMS. All other primary specifications are intended to be internally consistent.

5. Installation

DANGER — Disconnect and isolate hazardous energy before mounting or wiring.

Mounting

- 1 Install on a rigid, clean panel surface within the ambient range.
- 2 Maintain ventilation clearance.
- 3 Route wiring without terminal strain.
- 4 Protect signal wiring from abrasion and heat.

Sensor installation

- 1 Isolate the motor and ensure safe temperature.
- 2 Place TS1 at the designated measurement point.
- 3 Protect the cable.
- 4 Connect TS1 to TB2 with correct polarity.
- 5 Record sensor location.

CT1 installation

- 1 Confirm monitored conductor and CT ratio.
- 2 Install CT1 according to the fictional wiring arrangement.
- 3 Verify configured ratio.
- 4 Inspect CT connection.

5. Installation — Power-Up & Commissioning

WARNING — Confirm the motor cannot start unexpectedly before restoring control power.

Pre-power inspection

- TB1 secure
- TB2 secure
- CT1 installed and ratio recorded
- K1/K2 wiring checked
- J1/J2 seated
- Ventilation clear
- Covers/guards restored

Power-up

- 1 Apply control power without commanding motor operation.
- 2 Verify POWER and STATUS indications.
- 3 Check plausible temperature.
- 4 Verify CT ratio/current scaling.
- 5 Review alarm/trip thresholds.
- 6 Run under controlled conditions and observe temperature/current.
- 7 Record baseline values.

Commissioning observations

Check	Expected	If abnormal
POWER	Solid	Power diagnostics
Temperature	Plausible/stable	TS1 diagnostics
Current	Consistent with load	CT1/ratio diagnostics
Alarm relay	Operates at alarm condition	K1 circuit
Trip relay	Operates at trip condition	K2 circuit

6. Electrical Connections

DANGER — Electrical connections can expose personnel to hazardous voltage.

Terminal	Signal	Description
TB1-1	L1	Supply input
TB1-2	L2	Supply input
TB1-3	L3	Supply input
TB1-4	PE	Protective earth
TB2-1	TS1+	Temperature sensor positive
TB2-2	TS1–	Temperature sensor return
TB3-1	K1 COM	Alarm relay common
TB3-2	K1 NO	Alarm relay normally-open
TB3-3	K2 COM	Trip relay common
TB3-4	K2 NO	Trip relay normally-open
TB4-1	CT1+	Current transformer signal
TB4-2	CT1–	Current transformer return

Connection rules

- Do not assume an unlisted terminal is spare.
- Keep sensor/communication wiring protected.
- Verify relay contact ratings.
- Never connect supply conductors to sensor or CT terminals.

6. Electrical Connections — Signal Verification

TS1 sensor circuit

Condition	Expected behavior
Valid TS1	Temperature updates normally
Open circuit	TG-201; SENSOR flashes
Invalid range	TG-202
Intermittent	TG-203

CT1 current circuit

Condition	Expected behavior
Correct ratio	Reading agrees with verified reference
Wrong ratio	Consistent scaling error
Signal absent	Near-zero reading / TG-301
Loose connection	Fluctuation or intermittent zero

Relay circuits

K1 is the alarm relay and K2 is the trip relay. A local FAULT indication does not by itself prove that an external contactor changed state; the downstream control circuit must be checked separately.

CAUTION — Never force a relay output closed or bypass K2 to keep a motor operating.

6. Electrical Connections — Communication & Control

J2 provides a fictional RS-485 service interface. Communication settings include address, baud rate, parity and timeout.

Communication diagnostic order

- 1 Confirm stable controller power.
- 2 Confirm local indicator state.
- 3 Check J2 connector/cable.
- 4 Verify address and serial settings.
- 5 Review TG-401/TG-402.
- 6 Troubleshoot communication separately from primary thermal protection.

Diagnostic record

Item	Value
Motor operating current	___ A
TS1 temperature	___ °C
Alarm threshold	___ °C / ___ %
Trip threshold	___ °C / ___ %
CT ratio	___
Alarm delay	___ s
Trip delay	___ s
Error code(s)	___

7. Configuration

Parameter	Function
Temperature alarm	Starts alarm evaluation at threshold
Temperature trip	Initiates thermal protection after criteria persist
Overload threshold	Defines current level for overload evaluation
Alarm delay	Persistence time before alarm
Trip delay	Persistence time before trip
Reset mode	Defines reset behavior
Sensor type	Defines TS1 interpretation
CT ratio	Scales CT1 measurement
Communication settings	Defines J2 parameters

Configuration procedure

- 1 Record existing settings.
- 2 Confirm intended protection values from approved design.
- 3 Enter the value.
- 4 Save and verify retention.
- 5 Verify displayed values.
- 6 Run controlled test and record results.

NOTICE — Do not raise protection thresholds simply to prevent nuisance trips.

8. Operating Modes

Mode	Description
MONITOR	Normal temperature/current monitoring
ALARM	Alarm criterion satisfied; K1 active
TRIP	Protection trip; K2 active and FAULT indicated
RESET	Fault-clearing state
SERVICE	Configuration/diagnostic mode

9. Status Indicators

Indicator	Condition	Meaning
POWER	Solid	Controller powered
STATUS	Flashing	Normal monitoring
ALARM	Solid	Alarm active
FAULT	Solid	Protection trip/fault
SENSOR	Flashing	Sensor diagnostic condition
STATUS + FAULT	Alternating	Stored trip awaiting reset

10. Alarm and Fault Codes

All codes below are fictional TMS-450 identifiers.

Code	Message	Meaning	First check
TG-101	SUPPLY LOW	Supply below range	Supply/wiring
TG-102	SUPPLY HIGH	Supply above range	Source/wiring
TG-103	INTERNAL RESET	Unexpected restart	Supply/event history
TG-201	SENSOR OPEN	TS1 open circuit	TS1/TB2
TG-202	SENSOR RANGE	TS1 invalid range	Sensor/configuration
TG-203	SENSOR INTERMIT	TS1 intermittent	Cable/connector
TG-301	CURRENT LOST	CT1 signal absent	CT1/ratio
TG-302	CURRENT HIGH	Current above limit	Load/CT scaling
TG-401	COMM TIMEOUT	J2 timeout	Cable/settings
TG-402	COMM FRAME	Invalid frame	Cable/settings
TG-501	THERMAL TRIP	Thermal protection trip	Temp/load/cooling
TG-502	OVERLOAD TRIP	Overload trip	Current/threshold
TG-503	RESET BLOCK	Reset conditions unmet	Correct active condition
TG-504	RELAY CHECK	Relay diagnostic discrepancy	Relay wiring
TG-505	CONFIG ERROR	Invalid parameter	Configuration review

10. Fault-Code Interpretation

TG-501 THERMAL TRIP: protection operated. Possible contributors include elevated temperature, excessive load, persistent high current, restricted cooling, incorrect threshold, or inaccurate TS1 measurement.

TG-201 SENSOR OPEN: first inspect TS1, TB2 and the sensor circuit before considering replacement.

TG-503 RESET BLOCK: reset conditions remain unsatisfied. Correct the active condition before attempting reset.

WARNING — A protection fault may indicate a damaging condition. Do not defeat the protection function.

Evidence rule

A thermal trip proves that protection operated; it does not alone prove that the motor's actual temperature exceeded the threshold. Correlate code, temperature, current, cooling and configuration.

11. Troubleshooting — Problems 1–10

1. Motor temperature rises rapidly

Possible Causes: High load, restricted cooling, ambient heat, TS1 error

First Checks: Check load, ventilation, TS1 plausibility

Procedure: Correct verified load/cooling/sensor issue; do not simply raise threshold

Reference: §12, §14

2. Motor trips after ~10 minutes

Possible Causes: Persistent current, thermal accumulation, cooling, settings

First Checks: Record current and temperature at trip

Procedure: Follow thermal sequence and compare configuration

Reference: §12.3, §13

3. Thermal alarm but normal-looking temperature

Possible Causes: TS1/configuration/wiring

First Checks: Inspect TS1 and sensor type

Procedure: Verify measurement before changing thresholds

Reference: §14

4. Temperature reading unstable

Possible Causes: Loose connector, cable, interference

First Checks: Inspect TB2/cable

Procedure: Verify sensor circuit and trend

Reference: §14.4

5. Sensor fault appears

Possible Causes: Open/invalid TS1

First Checks: Inspect TS1/TB2

Procedure: Repair circuit; exact replacement part number not provided

Reference: §14

6. Current higher than expected

Possible Causes: Actual overload or CT scaling

First Checks: Compare independent current

Procedure: Verify load and CT ratio

Reference: §13, §15

7. Current is zero

Possible Causes: CT signal/wiring issue

First Checks: Inspect CT1

Procedure: Verify CT circuit

Reference: §15

8. Alarm relay does not activate

Possible Causes: K1/config/control circuit

First Checks: Check controller alarm state

Procedure: Separate controller output from downstream circuit

Reference: §11

9. Trip relay does not operate

Possible Causes: K2 wiring/condition

First Checks: Verify fault state

Procedure: Check K2 circuit

Reference: §18

10. Controller does not power on

Possible Causes: Supply/wiring

First Checks: Check supply safely

Procedure: Restore correct supply

Reference: §11.9

11. Troubleshooting — Problems 11–20

11. Status LED continuously flashes

Possible Causes: Normal STATUS flashing or abnormal pattern

First Checks: Identify exact LED/pattern

Procedure: Compare with Section 9 and code display

Reference: §9

12. Repeated startup trips

Possible Causes: High starting load, threshold/delay, sensor/cooling

First Checks: Record startup current/temperature

Procedure: Verify approved settings and startup behavior

Reference: §13

13. Thermal trip under normal load

Possible Causes: Sensor/cooling/ambient condition

First Checks: Compare current and temperature trend

Procedure: Inspect TS1, ventilation and load

Reference: §12–14

14. Temperature fixed at one value

Possible Causes: Sensor signal/processing

First Checks: Check whether value changes with conditions

Procedure: Inspect TS1 circuit

Reference: §14.4

15. Communication fault

Possible Causes: Cable/settings/interference

First Checks: Check J2/settings

Procedure: Verify communication parameters

Reference: §11

16. Reset does not clear fault

Possible Causes: Active condition/reset mode

First Checks: Read TG-503

Procedure: Correct condition, then reset

Reference: §11.6

17. Incorrect current displayed

Possible Causes: CT ratio/installation

First Checks: Verify known current and ratio

Procedure: Check CT1

Reference: §15.2

18. Alarm intermittent

Possible Causes: Threshold excursion/unstable signal

First Checks: Review event timing

Procedure: Correlate temp/current and wiring

Reference: §12, §14

19. Controller resets unexpectedly

Possible Causes: Supply instability/internal event

First Checks: Review TG-103

Procedure: Check supply and terminals

Reference: §11.9–11.10

20. Multiple alarms simultaneously

Possible Causes: Common supply/sensor/current/configuration issue

First Checks: Record all codes first

Procedure: Diagnose highest-risk condition then related signals

Reference: §10, §17

12. Thermal Protection

Thermal protection compares TS1 temperature with configured thresholds and timing. Thermal accumulation can be associated with sustained load, high current, restricted ventilation, elevated ambient temperature or a measurement problem.

Thermal alarm

When temperature reaches/exceeds the alarm threshold for the configured alarm delay, K1 alarm operation is initiated.

Thermal trip

When the trip criterion remains satisfied for the trip delay, K2 changes to the trip state and a protection fault is recorded.

12.3 Required diagnostic sequence

- 1 Stop the motor and isolate as required.
- 2 Record fault, temperature and current.
- 3 Check TS1 integrity and plausibility.
- 4 Verify motor load and current.
- 5 Inspect cooling/ventilation.
- 6 Compare thresholds/delays with approved design.
- 7 If current is high, investigate load.
- 8 If current is normal, investigate TS1 and cooling.
- 9 Record findings and restore only under approved procedure.

WARNING — Never bypass K2 or increase trip thresholds solely to prevent another trip.

13. Motor Overload Conditions

Excessive motor load may cause thermal accumulation. Persistent high current can result in thermal protection activation even when startup temperature appears normal.

13.1 Diagnostic relationship

Higher-than-expected current may indicate increased mechanical load, abnormal motor operation, incorrect CT scaling, or installation issues. Current must be correlated with temperature.

13.2 High-current investigation

- 1 Record displayed current.
- 2 Verify CT1 ratio and installation.
- 3 Obtain independent current measurement using approved method.
- 4 Compare with configured threshold.
- 5 Inspect driven load as permitted by its approved procedure.
- 6 If current is normal, continue with TS1/cooling diagnostics.

14. Temperature Sensor Troubleshooting

- 1 Isolate equipment.
- 2 Inspect TS1 mounting/cable.
- 3 Inspect TB2.
- 4 Confirm sensor type.
- 5 Check value stability/plausibility.
- 6 Use approved circuit test method.
- 7 Replace only through authorized procedure.

14. Sensor Troubleshooting — Detailed Tests

DANGER — Energized testing requires qualified personnel and applicable controls.

Symptom	Likely area	Action
TG-201	TS1 circuit	Inspect TB2, cable and sensor
TG-202	Sensor/configuration	Confirm sensor type and signal
TG-203	Connector/cable	Inspect strain and terminals
Fixed temperature	Sensor signal	Compare trend and inspect circuit
Low implausible alarm	Sensor scaling/type	Verify configuration

15. Current Measurement Troubleshooting

- 1 Record controller current and CT ratio.
- 2 Verify CT1 installation.
- 3 Inspect CT1 wiring.
- 4 Obtain independent current measurement.
- 5 Compare readings.
- 6 Correct approved ratio if wrong.
- 7 If ratio is correct, investigate CT wiring/circuit.

Multi-page evidence

A thermal-trip diagnosis requires the configuration evidence on page 12, thermal evidence on page 18, overload evidence on page 19, and sensor/current evidence on page 20.

15. Current Measurement — Worked Example

Example: the controller reports 74 A while expected operating current is approximately 52 A. Before concluding the motor is overloaded, verify CT1 ratio and obtain an independent measurement.

- If independent current is also high, investigate motor/load condition.
- If independent current is normal but the controller is high, verify CT ratio and installation.
- If the reading fluctuates, inspect CT wiring.
- If a thermal trip occurred, correlate current history with TS1 and cooling.

16. Controlled Test Scenario

Symptom: The motor operates normally at startup but stops after approximately 10 minutes and the ThermoGuard displays a thermal overload fault.

Recommended sequence: Record TG-501/TG-502 and temperature/current; check TS1; check motor load and current; inspect cooling; compare thresholds and delays; then correct the verified cause.

NOTICE — This scenario intentionally requires evidence from multiple sections/pages to test multi-chunk retrieval.

16. Preventive Maintenance

Interval	Inspection	Action
Before commissioning	Wiring/terminals	Verify secure and undamaged
Monthly	Enclosure/ventilation	Inspect for dust/blockage
Quarterly	TS1	Inspect mounting/cable
Quarterly	CT1	Inspect installation/terminals
Semiannual	Alarm function	Perform approved alarm test
Semiannual	Protection	Perform approved protection test
Annual	Configuration	Compare with approved design
Annual	Records	Review repeated faults/trends

17. Diagnostic Procedures

Use a hypothesis-driven sequence: establish symptom, collect evidence, isolate subsystem, verify, correct verified cause, and confirm normal operation. Avoid changing several parameters at once.

17. Diagnostic Procedures — Decision Logic

A. Thermal trip

- 1 Read code and record temperature/current.
- 2 Check TS1.
- 3 Check current/load.
- 4 Inspect cooling.
- 5 Review thresholds/delays.
- 6 Correct and verify.

B. Sensor fault

- 1 Record code.
- 2 Isolate.
- 3 Inspect TS1/cable.
- 4 Inspect TB2.
- 5 Verify sensor type.
- 6 Test/replace through approved procedure.

C. Current mismatch

- 1 Record current/ratio.
- 2 Verify CT1.
- 3 Measure independently.
- 4 Compare.
- 5 Correct configuration/installation if verified.
- 6 Repeat.

D. Multiple alarms

- 1 Record codes.
- 2 Check supply-related evidence.
- 3 Verify TS1/CT1.
- 4 Review configuration.
- 5 Investigate common causes.
- 6 Reset only after correction.

E. Communication fault

- 1 Record code.
- 2 Check J2.
- 3 Verify settings.
- 4 Inspect cable.
- 5 Confirm local protection.
- 6 Verify stable communication.

18. Replacement Procedures

DANGER — Replacement work requires energy isolation and verification.

Controller replacement

- 1 Record configuration/history.
- 2 Isolate and verify safe.
- 3 Label conductors.
- 4 Disconnect/remove controller.
- 5 Install approved replacement.
- 6 Reconnect according to terminal table/site drawing.
- 7 Restore approved configuration.
- 8 Perform commissioning checks.

19. Safety Procedures

WARNING — Do not bypass safety systems, defeat K2, short sensor inputs, or use temporary wiring to suppress protection faults.

Restoration checklist

- Tools/test equipment removed
- Covers/guards restored
- Terminals secured
- Configuration verified
- Protection not bypassed
- Personnel clear
- Authorized re-energization completed

Replacement sensor part number

The exact replacement sensor part number is intentionally not specified in this manual. The supported answer to a part-number question is therefore that the provided manual does not contain the exact ordering number.

20. Frequently Asked Technical Questions

Q: Why does the motor trip thermally?

A: Possible causes include excessive load, high current, restricted cooling, thermal conditions or inaccurate TS1 measurement.

Q: What should I check first after TG-501?

A: Record temperature/current and check TS1, load/current, cooling and configuration.

Q: What does TG-201 mean?

A: TS1 sensor open circuit.

Q: How identify a sensor fault?

A: Use TG-201/TG-202/TG-203 and inspect TS1 circuit/configuration.

Q: What causes unstable temperature?

A: Loose connections, cable damage, interference or intermittent sensor.

Q: What causes excessive current?

A: Increased load, CT scaling/installation issues or abnormal motor operation are supported possibilities.

Q: When should K1 activate?

A: When the alarm criterion persists for the configured alarm delay.

Q: What after a thermal trip?

A: Isolate as required, record evidence, diagnose, correct and verify.

Q: Can I reset immediately?

A: Not as a substitute for correcting the active condition.

Q: What causes repeated thermal trips?

A: Persistent load/current, cooling restrictions, settings or measurement problems.

Q: Maximum monitored current?

A: 80 A RMS in the primary specification table.

Q: Default trip temperature?

A: 110 °C.

Q: Supply range?

A: 100–240 VAC, 50/60 Hz.

Q: Sensor range?

A: –20 °C to +150 °C.

Q: Default CT ratio?

A: 20:1.

21. Troubleshooting Decision Tables

Condition	First decision	Next evidence
TG-501	Is temperature genuinely elevated?	TS1 + trend
TG-501 + normal temperature	Is current/load elevated?	CT1/current
High current + high temperature	Is load above expected?	Load inspection
Normal current + high temperature	Is cooling adequate?	Ventilation + TS1
High displayed current + normal independent current	Is CT ratio correct?	CT1/configuration
TG-201	Is TS1 circuit open?	TB2/cable/sensor
TG-401	Is J2 needed for primary protection?	Local protection + comm checks

22. Appendix — Reference Data

The following values are repeated for convenience. One controlled documentation inconsistency is intentional.

Parameter	Appendix value
Supply	100–240 VAC, 50/60 Hz
Maximum monitored current	75 A RMS
Temperature sensor range	–20 °C to +150 °C
Default alarm	95 °C
Default trip	110 °C
K1/K2 rating	250 VAC, 3 A resistive maximum

NOTICE — The Appendix maximum monitored current of 75 A RMS conflicts with the primary Technical Specifications value of 80 A RMS on page 6. Do not silently reconcile the discrepancy.

Unsupported information boundary

This manual does not document underwater operation, aircraft, automotive, solar panels, medical equipment, nuclear equipment, specific competitors, real manufacturer products, or real-world certification claims.

23. Specifications Summary

Item	Primary documented value	Page
Supply	100–240 VAC, 50/60 Hz	6
Maximum monitored current	80 A RMS	6
Temperature range	–20 °C to +150 °C	6
Alarm	95 °C default	6
Trip	110 °C default	6
Operating ambient	–20 °C to +55 °C	6
Relay rating	250 VAC, 3 A resistive maximum	6
Digital input	24 VDC	6
Communication	J2 fictional RS-485	6
Enclosure	IP42	6
Weight	0.62 kg	6
Dimensions	145 × 95 × 62 mm	6

RAG testing coverage

- Exact retrieval: specifications, terminals, codes.
- Semantic retrieval: overheating, thermal rise, elevated temperature, excessive load.
- Multi-page evidence: 10-minute thermal overload scenario.
- Partial evidence: replacement sensor part number absent.
- Unsupported questions: excluded domains.
- Conflict detection: 80 A versus 75 A.
- Safety retrieval: warnings near procedures.

NOTICE — Fictional manual for FixRAG evaluation only. Not a real product manual or certified safety procedure.